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Lectures

1.1 The Roles of Early Keyboard Instruments

1.1.1 Mid-17th Century Germany

Historical Background

In the mid-17th century, Germany experienced major political and social changes that strongly influenced the development of music, especially keyboard music.

The **Treaties of Westphalia (1648)** marked the end of the Thirty Years' War¹. Although peace was restored, the treaties left the Germanic lands politically fragmented rather than unified.

Political Structure

After 1648, the Germanic territories were divided into hundreds of small states, such as duchies, bishoprics, and free cities². Each region was ruled by local noble authorities who maintained their own courts and cultural institutions.

This political fragmentation encouraged regional musical styles and led to strong traditions of court-sponsored music, including keyboard performance and composition.

Social Strata

German society at the time was clearly divided into three major social classes:

- **Noble lords:** The aristocracy who controlled land, wealth, and political power. They were the primary patrons of musicians and owned expensive instruments such as harpsichords.
- **Burghers:** The urban middle class, including merchants and educated professionals³. They increasingly valued music education and domestic music-making.

¹The Thirty Years' War (1618–1648) was a devastating religious and political conflict in Europe that caused massive population loss and economic destruction, particularly in German lands.

²A free city was a city that enjoyed political autonomy under the Holy Roman Empire.

³Burghers were town citizens who were economically independent but did not belong to the nobility.

- **Peasants:** The rural population with limited access to formal education or musical training.

Access to music education and instruments was largely determined by social class.

Language and Intellectual Culture

Latin was the dominant language of education, religion, and intellectual discourse⁴. Music theory treatises and educational materials were commonly written in Latin, reinforcing the idea that music was an intellectual and scholarly discipline.

Rationalism and Doctrinal Purity

The intellectual climate was shaped by **rationalism**, which emphasized reason, order, and logical structure⁵. In Protestant regions of Germany, there was also a strong focus on doctrinal purity, clarity, and discipline in both religion and art.

These values influenced music by encouraging structural clarity, controlled expression, and systematic keyboard training, laying the foundation for later developments in keyboard and piano music.

1.1.2 Musical and Social Scenario

The musical culture of mid-17th century Germany reflected the social hierarchy and intellectual values of the time.

Music was regarded primarily as an **intellectual activity rather than an emotional one**⁶. As a result, composers focused on the **development of musical motives**, emphasizing systematic variation and formal coherence.

Musical practices varied significantly across social classes:

- **Peasants** primarily engaged in simple songs and dance music, which were transmitted orally and served social or festive functions.
- **Princes** maintained chamber orchestras to perform instrumental and court music within palaces.
- **Kings and dukes** supported large-scale opera productions, which required significant financial and organizational resources⁷.

⁴Latin functioned as a universal scholarly language across Europe, symbolizing education and authority.

⁵Rationalism is a philosophical movement that values reason as the primary source of knowledge.

⁶This reflects the influence of rationalism, which valued structure, logic, and control over spontaneous emotional expression.

⁷Opera was an elite art form involving singers, orchestra, stage design, and theater infrastructure.

Music also reflected a strong sense of **foreign influence**. Italian opera was admired for its vocal style and dramatic expression, while court dances often followed French stylistic models⁸.

The **educated burgher class** favored sacred music such as chorales and cantatas, particularly within the church setting. These genres combined religious devotion with intellectual musical structure and were accessible to the literate middle class.

1.1.3 The Organ

The **organ** was regarded as the “**king of instruments**” during the mid-17th century due to its grandeur, complexity, and intellectual status⁹. It was considered lofty, intricate, and highly intellectual, and was primarily used in churches.

The organ played a central role in **polyphonic music**¹⁰, making it especially suitable for complex contrapuntal compositions. Because of its size and cost, the organ was not a domestic instrument but a permanent fixture in sacred spaces.

Principles of the Organ

The organ produces sound through the controlled flow of air:

- Sound is generated when **air flows through valves** and into pipes.
- **Different pitches** are produced by pipes of different lengths¹¹.

Tone Control and Dynamics

The **tone quality** of the organ is controlled using **stops**, which activate different sets of pipes¹². Stops can be combined to create rich and complex sounds.

Dynamic changes on the organ are limited but achievable through several methods:

- **Swell mechanisms**, which enclose pipes in a box with movable shutters.
- Use of **different manuals** (keyboards), each controlling different pipe divisions.
- **Holding keys** to sustain sound, as organ tones do not decay naturally.

⁸France was seen as a cultural leader in court etiquette, dance, and artistic taste during this period.

⁹The phrase “king of instruments” reflects the organ’s large scale, wide pitch range, and strong association with sacred and intellectual traditions.

¹⁰Polyphonic music consists of multiple independent melodic lines sounding simultaneously, requiring precise control and clarity.

¹¹Longer pipes produce lower pitches, while shorter pipes produce higher pitches.

¹²Each stop corresponds to a particular timbre or pipe group, allowing varied tone colors.

Sound Characteristics, Capabilities, and Limitations

The organ is characterized by a **rich, brilliant, and majestic sound**, making it ideal for large architectural spaces such as churches.

Its primary strength lies in its ability to perform **complex polyphonic music**, supporting multiple independent voices simultaneously. However, the organ has limitations in expressive dynamics and articulation compared to later keyboard instruments, influencing the compositional styles written for it.

1.1.4 The Harpsichord

The **harpsichord** was a dominant keyboard instrument in the 17th century, especially in courtly and theatrical settings. It appeared under different names and forms across Europe:

- **Clavicembalo** (Italian)
- **Clavecin** (French)
- **Flügel** (German)
- **Spinet**: a smaller harpsichord with strings arranged at an angle to the keyboard¹³
- **Virginal**: a simple rectangular keyboard instrument¹⁴

The harpsichord was regarded as **luxurious and elegant**, and was primarily associated with **privileged aristocrats**. It was widely used in **opera productions** and **orchestral settings**, as well as for solo repertoire and basso continuo.

Principles of the Harpsichord

- **Mechanism of sound production**: When a key is pressed, a mechanical action causes a small plectrum to **pluck a string**¹⁵, producing sound. This plucking mechanism distinguishes the harpsichord from the organ, which relies on airflow.
- **Different pitches**: Pitch is determined by the **length, thickness, and tension of the strings**, similar to other string instruments.
- **Different dynamics**: The harpsichord has **very limited dynamic control**¹⁶. Pressing a key harder does not significantly change the loudness. Dynamic contrast is achieved mainly through:
 - Use of multiple manuals (if available)

¹³Spinet (短翼羽管键琴): 体积较小, 琴弦与键盘呈斜角排列。

¹⁴Virginal (维吉纳琴): 矩形外形, 结构简化, 常用于家庭。

¹⁵Plectrum (拨片): 通常由羽毛或皮革制成, 用于拨动琴弦。

¹⁶Dynamic control (力度控制): 指通过触键力度改变音量的能力。

- Changing registers or stops
- **Tone quality:** The tone is **bright, crisp, and clear**, with little sustain. This makes the instrument especially suitable for rhythmically precise and highly articulated music.

Sound Characteristics, Capability, and Limitation

The harpsichord is characterized by a **brilliant and percussive sound**, with rapid decay after each note. It excels in:

- Fast passages
- Ornamentation
- Polyphonic textures¹⁷

However, its primary limitation is the **lack of expressive dynamic variation**, which restricts emotional contrast. This limitation played a crucial role in motivating the later development of the piano, an instrument capable of responding to touch-sensitive dynamics.

1.1.5 The Clavichord

The **clavichord** was a simple and intimate keyboard instrument, widely used for domestic music-making, especially in German-speaking regions. It was inexpensive, compact, and commonly found in middle-class households, making it a typical **German homemade product** rather than a court instrument.

Principles of the Clavichord

- **Mechanism of sound production:** The clavichord is **stroke-responsive**, meaning the sound is directly controlled by the player's touch. When a key is pressed, a small metal **tangent** strikes the string and remains in contact with it¹⁸. Unlike the harpsichord, the tangent both initiates and sustains the vibration.

The mechanism is extremely simple: the **key and tangent form a single piece**, and there are **no stops or registers**, reflecting the instrument's modest design.

- **Different pitches:** Two structural types of clavichord existed:
 - **Fretted clavichord:** one string produces two or more different pitches¹⁹.
 - **Unfretted clavichord:** each pitch has its own string, resulting in a larger instrument with clearer intonation.

¹⁷Polyphonic texture (复调织体): 多条独立旋律同时进行的音乐结构。

¹⁸Tangent (切弦片): 固定在琴键末端的金属片, 用于敲击并压住琴弦。

¹⁹Fretted (共弦式): 多个音高共用一根琴弦, 通过不同触键位置改变音高。

- **Different dynamics:** The clavichord allows **true dynamic control**, as pressing the key harder increases volume. It also supports **Bebung**, a vibrato-like expressive effect created by varying finger pressure while holding a key²⁰.
- **Tone quality:** The tone is **soft, delicate, and intimate**, with very low volume, making it unsuitable for large spaces or ensemble performance.

Sound Characteristics, Capability, and Limitation

The clavichord is characterized by its **high expressive sensitivity** and close connection between finger and sound. It excels as:

- A practice instrument
- A tool for musical education
- An expressive medium for private performance

However, its primary limitation is its **extremely low volume**, which restricts its use to small rooms and solo contexts. Despite this limitation, the clavichord played a crucial role in shaping expressive keyboard technique and directly influenced later developments in piano touch and articulation.

1.1.6 Keyboard Business

During the mid-17th century, keyboard instrument making was not yet a fully independent industry. Instead, it functioned largely as a **sideline business** for **organ builders** or **cabinet makers**²¹. These craftsmen already possessed the woodworking and mechanical skills required to construct keyboard instruments.

Demand, Production, and Sale

The demand for keyboard instruments grew alongside the expansion of court music and domestic music-making. Production was typically small-scale and customized rather than standardized. In many cases, instruments were produced and sold **by delivery**²², reflecting the artisanal nature of the trade.

Terminology and Historical Usage

Terminology related to keyboard instruments evolved over time:

²⁰Bebung (颤音效果): 通过持续按压并微调压力产生的音高与音量变化, 是羽管键琴无法实现的表现手法。

²¹Cabinet makers (家具木匠) 具备制作精细木制外壳的技能, 非常适合键盘乐器的结构需求。

²²Delivery (定制交付): 根据客户需求制作并直接交付, 而非大规模库存销售。

- **Clavier**: originally referred to string keyboard instruments such as the **clavichord** or **harpsichord**²³. Later, the term came to include the **pianoforte**.
- **Flügel**: originally used to describe a **harpsichord**, especially wing-shaped instruments²⁴. Over time, the term evolved to refer to the **grand piano**.

These shifting terms reflect both technological development and changing musical preferences, marking the transition from early string keyboard instruments toward the piano.

1.2 Happy Birthday, Piano!

1.2.1 Early 18th Century Germany: Context and Culture

The Rise of Pietism

In the early 18th century, Germany experienced a significant religious and cultural shift known as **Pietism**²⁵.

- **Core Principles**: The movement sought to combine **Lutheran**²⁶ principles with individual piety.
- **Personal Devotion**: There was a shift from rigid religious structure to being a "religious enthusiast." The emphasis was placed heavily on **personal devotion** rather than public ritual.
- **Emotion over Dogma**: Pietism prioritized "feeling" over **doctrine**²⁷. The internal emotional state of the believer was more important than theological complexity.
- **Demographics**: This movement was particularly favored by the "unlearned" (common people) and women, as it was accessible and personal.
- **Social Impact**: It was not a political revolution, but a spiritual one. It encouraged the usage of the **native German language** in worship and daily life, replacing the traditional Latin used in formal church settings.

Musical Implications of Pietism

The cultural shift towards Pietism directly influenced musical tastes and instrument preferences during this period.

²³Clavier (键盘弦乐器的统称): 早期并不特指某一种具体乐器。

²⁴Flügel (翼形琴): 名称源于乐器外形。

²⁵Pietism (虔敬主义): A movement within Lutheranism that combines its biblical principles with the Reformed emphasis on individual piety and living a vigorous Christian life.

²⁶Lutheran (路德宗): A major branch of Protestant Christianity which identifies with the theology of Martin Luther.

²⁷Doctrine (教条/教义): A belief or set of beliefs held and taught by a church, political party, or other group.

- **Aesthetic Preference:** There was a strong favor for **musical simplicity** and a rejection of excessive complexity (such as dense polyphony). Music was meant to speak directly to the heart.
- **The Clavichord:** The **Clavichord**²⁸ was the preferred instrument of this era because:
 - It offered control over sound intensity (dynamics).
 - It allowed for a better connection to "feeling" and personal expression.
- **Arte Modulatoria:** This concept refers to the "Art of Singing" on an instrument. It emphasizes the **shading of sounds**²⁹, aiming to make the keyboard instrument "sing" with human-like nuance.

Musical Simplicity: Johann Mattheson's Perspective (1739)

This section highlights the aesthetic shift towards simplicity in the 18th century, as articulated by the music theorist **Johann Mattheson**.

- **Preference for Melody:** Mattheson argues that the ear finds more delight in a "single well-ordered voice" carrying a **clean melody**.
- **Critique of Complexity:** He criticizes complex musical textures (metaphorically referring to "four-and-twenty" voices), stating that the musical meaning gets "torn to pieces" when distributed over too many parts. This is a rejection of dense **polyphony**³⁰.
- **The Power of Simplicity:** He advocates for "unadorned melody"³¹ in its "noble simplicity, clarity, and distinctness".
- **Emotional Connection:** The ultimate goal is to "move hearts." Mattheson believes that natural simplicity often surpasses the "arts of **harmony**"³² in achieving this emotional impact.

²⁸Clavichord (击弦古钢琴): An early keyboard instrument that produces sound by striking strings with small metal blades (tangents). It is known for its ability to produce dynamic changes (volume) and vibrato (Bebung), unlike the harpsichord.

²⁹Shading of sounds (音色的明暗/细微变化): The subtle graduation of tone and volume to create expression, similar to how a vocalist shapes a phrase.

³⁰Polyphony (复调音乐): A style of musical texture consisting of two or more simultaneous lines of independent melody, as opposed to a single melody with accompaniment. Mattheson's "four-and-twenty" is an exaggeration of this complexity.

³¹Unadorned melody (朴实无华的旋律): A melody that is simple, natural, and free from excessive ornamentation or decorative notes.

³²Harmony (和声): The combination of simultaneously sounded musical notes to produce chords and chord progressions having a pleasing effect. Mattheson argues here that melody is superior to complex harmony for expressing emotion.

Favor Clavichord: Mattheson's Comparison (1713)

- **Ideal Genres:** Mattheson argued that genres such as **Overtures**, **Sonatas**, **Toccatas**, and **Suites** are "brought out the best and the cleanest" on a good clavichord.
- **Expressive Capabilities:**
 - The instrument allows the player to express the **art of singing** (cantabile style) much more intelligibly.
 - It is capable of "sustaining and softening" notes, offering dynamic nuance.
- **Critique of Plucked Instruments:** Mattheson contrasted the clavichord with the **Harpsichord**³³ and **Spinnet**³⁴. He described them as being "always alike strongly resonant," implying a lack of variety in volume and tone.
- **Test of Skill:** The clavichord was seen as the true test of a musician's touch. Mattheson advised that to hear a "delicate hand and a pure style," one should lead the candidate to a "neat clavichord".

Pantaleon Hebenstreit and the Pantalon

This section discusses Pantaleon Hebenstreit, a figure whose instrument bridged the gap between struck string instruments and the keyboard.

- **The Musician:** Pantaleon Hebenstreit was a renowned **Dulcimer**³⁵ player from Leipzig.
- **The Instrument's Origins:** The instrument he played was essentially a huge **Cimbalom**³⁶.
 - At the time, this instrument was associated with the lower classes and used in puppet shows.
 - In German, it was known as the **Hackbrett**³⁷, implying a crude method of playing.
- **Musical Innovation:** Despite its humble origins, Hebenstreit demonstrated that the instrument was "stroke responsive." This allowed for a wide variety of **nuances**³⁸

³³Harpsichord (大键琴/羽管键琴): A keyboard instrument where strings are plucked. It produces a loud, brilliant sound but generally lacks the ability to make gradual dynamic changes (loud/soft) simply by touch.

³⁴Spinnet (小型大键琴): A smaller, usually wing-shaped or pentagonal version of the harpsichord, also producing sound by plucking strings.

³⁵Dulcimer (扬琴): A stringed instrument with a trapezoidal soundboard, played by striking the strings with small hammers.

³⁶Cimbalom (辛巴龙): A large, chromatic concert dulcimer, historically popular in Gypsy music (Romani music) in Eastern Europe.

³⁷Hackbrett (德语 "剁肉板" /扬琴): Literally translating to "chopping board" or "hacking board," referring to the striking motion used to play it.

³⁸Nuance (细微差别): Subtle differences in expression, volume, or tone color.

depending on how hard the strings were struck.

- **Royal Recognition:** King Louis XIV³⁹ was so impressed that he renamed the instrument the "Pantalon" (or Pantaleon) after the player.
- **Limitations and Evolution:**
 - The Pantalon was cumbersome (difficult to handle) and expensive to maintain.
 - This difficulty led to a critical design question that paved the way for the piano: "Why not fitted to finger-playable keys?"

1.2.2 Bartolomeo Cristofori's New Invention (1700)

This section introduces the inventor of the piano and the first historical documentations of the instrument.

- **The Inventor and Patron:** The instrument was invented by **Bartolomeo Cristofori**⁴⁰, who worked for **Prince Ferdinand de' Medici**⁴¹ in Florence.
- **First Historical Record (1700):**
 - The first record appears in the Medici inventory by Federigo Meccoli.
 - It describes the instrument as an "Arpicembalo del piano e forte" (Harpsichord with soft and loud), invented by Cristofori in the year 1700.
- **Documentation by Scipione Maffei (1711):**
 - The invention was detailed in the *Journal of Letters of Italy* by **Scipione Maffei**⁴².
 - Maffei described it as a "new invention of a harpsichord (gravicembalo) with soft and loud".
 - The article included a diagram of the mechanism and described successful concerts held in Rome.
- **Musical Capabilities:** Maffei highlighted specific musical virtues of the new instrument:
 - **Dialogue Manner:** It allowed for a "dialogue manner" with propositions and responses between phrases.

³⁹Louis XIV (路易十四): The King of France (1643–1715), known as the "Sun King," a major patron of the arts.

⁴⁰Bartolomeo Cristofori (巴托洛梅奥·克里斯托福里): An Italian instrument maker from Padua, credited with inventing the piano.

⁴¹Prince Ferdinand de' Medici (费迪南多·德·美第奇王子): A Grand Prince of Tuscany who was a passionate patron of music and employed Cristofori in Florence.

⁴²Scipione Maffei (希皮奥内·马费伊): An Italian writer and art critic whose 1711 article provided the first detailed description and diagram of the piano mechanism.

- **Dynamic Contrast:** It was capable of contrast and **gradual change of volume**⁴³ within phrases.

Scipione Maffei's Account and Evaluation (1711)

This section details Scipione Maffei's 1711 article in the *Journal of Letters of Italy*, which provides the rationale for the piano's invention and addresses early criticisms.

- **The Musical Necessity:** Maffei argues that the "principal source" of musical delight is the **alternation of soft and loud**⁴⁴. This includes:
 - Sudden changes (between a "thesis and its response").
 - Gradual changes (diminishing little by little, or returning to "full vigor").
- **The Limitation of Existing Keyboards:** While **bowed instruments**⁴⁵ excel at this diversity of tone, the traditional harpsichord is "entirely deprived" of this capability.
- **Cristofori's Solution:** Bartolomeo Cristofori (referred to as "Ciristofali") constructed an instrument where sound volume depends on the **varying force**⁴⁶ with which keys are pressed. This allows for gradations and diversity of sound, comparable to a **violoncello**⁴⁷.
- **Early Criticisms:** Maffei notes that some "professors" did not initially approve of the instrument:
 - They failed to understand the **ingenuity**⁴⁸ required to overcome the mechanical challenges.
 - They felt the tone was "muffled and dull" compared to the usual "silvery sound" of harpsichords.
- **Maffei's Defense:** He argues that this perception is merely due to **habituation**⁴⁹. He claims the ear adapts quickly and will eventually take "no more pleasure in the ordinary harpsichord".

⁴³Gradual change of volume (音量的渐变): The ability to increase or decrease volume smoothly (crescendo/diminuendo) within a phrase, which was impossible on a standard harpsichord.

⁴⁴Alternation of soft and loud (强弱交替): The contrast between dynamic levels, which provides musical expression and emotional depth.

⁴⁵Bowed instruments (弓弦乐器): Instruments like the violin or cello where sound is produced by drawing a bow across strings, allowing for continuous dynamic control.

⁴⁶Varying force (不同的力度): The physical principle of the piano action where a harder key press results in a louder sound, and a softer press results in a quieter sound.

⁴⁷Violoncello (大提琴): A bowed string instrument known for its wide dynamic range and expressive, singing tone.

⁴⁸Ingenuity (独创性/精巧): The quality of being clever, original, and inventive.

⁴⁹Habituation (习惯化): The diminishing of a physiological or emotional response to a frequently repeated stimulus. Here, it means listeners were just too used to the bright sound of the harpsichord.

Evaluating Cristofori's Piano: Addressing Criticisms

This section continues Scipione Maffei's defense of the instrument, specifically addressing the criticism regarding its volume and defining its proper musical context.

- **The Criticism:** A common complaint was that the instrument did "not have enough volume of tone".
- **Rebuttal 1 - Technique:** Maffei argues that the instrument actually possesses significantly more tone than people realize, provided the player knows how to bring it out by "striking the keys with **vigor**"⁵⁰.
- **Rebuttal 2 - Proper Context:**
 - One must judge the instrument based on its intended purpose.
 - It is properly a **Chamber Instrument**⁵¹.
 - It is *not* suitable for **Church Music**⁵² or for a "large orchestra".
- **Ideal Usage:** The piano succeeds perfectly for:
 - Accompanying a singer.
 - An ensemble of a moderate number of pieces.
 - Playing alone, similar to the **Lute** or **Harp**.
- **Demands on the Player:** As a new instrument, it requires a player who has made a "particular study" of it. The player must:
 - Understand its virtues.
 - Regulate the **impulse**⁵³ imparted to the keys.
 - Achieve "graceful gradations" at the right time and place.

Cristofori's Piano Action Mechanism

This section details the mechanical innovation that allowed the piano to function, known as the "Action."

- **Core Components:** Cristofori's mechanism involved several key innovations distinct from the harpsichord or clavichord:

⁵⁰Vigor (活力/力度): Physical strength and energy; here referring to the forceful touch required to produce a loud sound on the piano.

⁵¹Chamber instrument (室内乐乐器): An instrument intended for performance in a smaller room (chamber) rather than a large hall, often for intimate music-making.

⁵²Church music (教会音乐): Music intended for performance in a church, typically requiring instruments like the organ that can fill a vast, resonant space.

⁵³Impulse (冲力/脉冲): The force or driving motion applied to the keys.

- Hammers swing from a separate rail or rack.
- The inclusion of an **Escapement**⁵⁴.
- A **Synchronous Damper**⁵⁵ and a "check."
- **The "Toss" Principle:**
 - Unlike the clavichord (tangent stays in contact) or harpsichord (pluck), the piano action is designed to "**Toss the hammer**". The hammer is thrown at the string and then falls back.
 - If the key is pressed too slowly, there is "no toss," meaning the hammer might not reach the string with sufficient force to create sound (or simply describes the separation of the hammer from the key motion).
 - This allows for the **gradual control** of the hammer towards the string, directly translating finger speed to hammer speed (and thus volume).
- **The Check (Back Catch):**
 - The mechanism includes a **Check** or **Back Catch**⁵⁶.
 - Its purpose is to prevent the "hammer re-rebound," ensuring a clean single strike.
- **Key Structural Parts (Diagrammatic):**
 - **Jack**⁵⁷: Transfers energy from the key to the hammer.
 - **Hammer Shank**⁵⁸: Supports the hammer head.
 - **Knuckle**⁵⁹: The contact point for the jack.
 - **Wippen Assembly**⁶⁰: The complex lever system facilitating the action.
 - **Capstan**⁶¹: Adjusts the regulation between the key and the action.

⁵⁴Escapement (擒纵结构): A mechanism that allows the hammer to escape from the jack (the part pushing it) just before it hits the string. This ensures the hammer strikes the string and immediately rebounds, letting the string vibrate freely even if the key is still held down.

⁵⁵Synchronous Damper (同步制音器): A felt pad that lifts off the string when the key is pressed (allowing sound) and falls back onto the string when the key is released (stopping sound).

⁵⁶Check/Back Catch (制动器/托木): A device that catches the hammer on its rebound from the string to prevent it from bouncing back up and hitting the string a second time (re-rebound) while the key is still depressed.

⁵⁷Jack (顶杆): The lever that pushes the hammer assembly upward when the key is pressed.

⁵⁸Hammer Shank (槌柄): The thin rod that connects the hammer head to the mechanism.

⁵⁹Knuckle (关节/轴): A small, leather-covered cylinder attached to the hammer shank that the jack pushes against.

⁶⁰Wippen Assembly (韦佩结构/联动杠杆): The intermediate lever system that sits between the key and the hammer, holding the jack and other components.

⁶¹Capstan (卡钉): A screw used to adjust the height and contact between the key and the wippen.

The Italian Stagnation: Questions to Ponder

Despite the innovation, the piano did not immediately dominate the Italian musical landscape. This section explores why the Italians did not continue to heavily develop the pianoforte.

- **Commercial Failures:**
 - There was no concept of **Patenting**⁶² the invention for profit.
 - There was no organized production or marketing strategy.
- **Cultural Context:**
 - **Opera Dominance:** Italy had been the home of **Opera**⁶³ since the late 16th century.
 - **Violin Supremacy:** The period of 1700–1725 was the "Golden Period" of **Stradivari**⁶⁴, further cementing the dominance of string instruments.
- **Luxury Status:** The early pianos were considered a "luxurious furnish" (furniture), intended for aristocrats like the Medici, not for mass production or general public use.

Fortepiano Capability Summary

This section summarizes the unique musical capabilities of Cristofori's fortepiano compared to its predecessors.

- **Touch Responsive Mechanism:** The key innovation provided:
 - Volume contrast and graduation in sound.
 - Control over the **Decay**⁶⁵ of sound.
- **Stylistic Suitability:**
 - **Singing Style:** It achieved a "singing style quality" (cantabile).

⁶²Patent (专利): A government authority or license conferring a right or title for a set period, especially the sole right to exclude others from making, using, or selling an invention. Cristofori did not patent his invention to make a profit.

⁶³Opera (歌剧): A dramatic work in one or more acts, set to music for singers and instrumentalists. It was the dominant form of entertainment in Italy since the late 16th century, overshadowing instrumental innovations.

⁶⁴Stradivari (安东尼奥·斯特拉迪瓦里): A renowned Italian luthier (stringed instrument maker) whose violins are considered the finest ever made. The focus on violin perfection may have distracted from keyboard development.

⁶⁵Decay (声音的衰减): The way a sound fades into silence. On a piano, the player can control how long a note rings before the damper stops it.

- **Baroque Style:** It was well-suited for the **Baroque**⁶⁶ musical style, specifically the concept of "Proposition and Response" (dialogue between phrases).
- **Melody:** It was capable of bringing out the melody against the accompaniment.
- **Acoustic Positioning:**
 - It was louder than the clavichord.
 - It remained a **Chamber Instrument** (for small rooms).
 - It could **imitate the orchestra**, but was not meant to be played *in* the orchestra (at this stage).

Cristofori's Influences and Early Repertoire

This section discusses the first music written specifically for the piano and how the instrument began to spread beyond Italy.

- **Ludovico Giustini's Sonatas (1732):**
 - **Ludovico Giustini**⁶⁷ published twelve **Sonatas**⁶⁸.
 - This was the **first publication** for the "piano and forte keyboard, commonly called hammer," appearing in 1732.
 - The work was dedicated to the younger brother of the **Portuguese King**.
 - **Structure:** The pieces contained four or five movements written in traditional **dance styles**⁶⁹.
 - **Innovation:** Crucially, these scores included specific **dynamic markings**⁷⁰.
- **Iberian Connection:**
 - Five Florentine pianos were sold to the royal families in **Portugal and Spain**.
 - This is historically significant because the famous composer **Domenico Scarlatti**⁷¹ worked there, suggesting he had access to these early pianos.
- **Spread to Germany:**

⁶⁶Baroque (巴洛克): A style of European architecture, music, and art of the 17th and 18th centuries, characterized by ornate detail. In music, it often involves a "Proposition and Response" structure.

⁶⁷Ludovico Giustini (路多维科·朱斯蒂尼): An Italian composer credited with writing the first music specifically for the piano.

⁶⁸Sonata (奏鸣曲): A musical composition for a solo instrument, usually consisting of several movements.

⁶⁹Dance styles (舞曲风格): Musical forms based on dances (like the Allemande, Courante, Sarabande), common in the Baroque era.

⁷⁰Dynamic markings (力度记号): Symbols in sheet music indicating how loud or soft to play (e.g., *p* for piano/soft, *f* for forte/loud), which were necessary to utilize the piano's unique capabilities.

⁷¹Domenico Scarlatti (多梅尼科·斯卡拉蒂): An Italian composer who spent much of his life in the service of the Portuguese and Spanish royal families. He is famous for his 555 keyboard sonatas.

- Knowledge of the instrument spread to Germany via an essay by Johanne Ulrich von Koenig, which appeared in **Johann Mattheson's** *Critica Musica* in 1725.
- **J.S. Bach and Frederick the Great (1747)**: A famous historical event occurred when J.S. Bach⁷² played on a piano made by **Gottfried Silbermann**⁷³ before **Frederick the Great**⁷⁴ at the Prussian Court in 1747.

1.3 Piano Growing Popularity

1.3.1 Cristofori's Influences

The First Published Piano Music

- **Ludovico Giustini's Twelve Sonatas (1732)**: These represent the earliest known music specifically published for the piano.
- **Original Title**: *Sonate da cimbalo di piano e forte detto volgarmente di martelletti*⁷⁵.
- **Translation**: "Sonata for harpsichord of piano and forte, commonly called hammer".
- **Royal Dedication**: The work was dedicated to the younger brother of the King of Portugal.

Technical and Regional Dissemination

- **Dynamic Markings**: The sonatas included specific dynamic markings⁷⁶, which were a novelty compared to traditional harpsichord repertoire.
- **Iberian Influence**: Florentine pianos⁷⁷ were sold to the royal families in Portugal and Spain.
- **Domenico Scarlatti**: These instruments reached the courts where D. Scarlatti was employed, influencing the keyboard culture of the region.

⁷²J.S. Bach (约翰·塞巴斯蒂安·巴赫): A magnificent German composer and musician of the Baroque period.

⁷³Gottfried Silbermann (哥特弗里德·希尔伯曼): A German keyboard instrument maker who built early pianos based on Cristofori's designs.

⁷⁴Frederick the Great (腓特烈大帝): King of Prussia, a military tactician and a flautist/composer who patronized the arts.

⁷⁵意为“拨弦琴上的强弱音奏鸣曲，通常被称为小锤子（钢琴）”。

⁷⁶强弱记号

⁷⁷佛罗伦萨钢琴

1.3.2 Germany Before 1750

Early Dissemination and Translation

- **Maffei's Influence:** J.U. Koenig translated Scipione Maffei's article (which described Cristofori's invention) into German.
- **Publication:** It was published in 1725 within Johann Mattheson's *Critica Musica*.

Gottfried Silbermann (1683–1753)

- **Background:** A renowned organ builder based in Freiberg.
- **Technical Roots:** He was responsible for maintaining Hebenstreit's dulcimer⁷⁸, which influenced his understanding of hammer-based string instruments.
- **C.P.E. Bach's View:** While Silbermann is famous for pianos, C.P.E. Bach specifically praised his clavichords⁷⁹.

The Interaction with J.S. Bach

- **Initial Reaction (1736):** J.S. Bach was famously unimpressed with Silbermann's early fortepiano⁸⁰, criticizing its heavy touch and weak treble.
- **Later Approval (1747):** Bach eventually approved of the fortepianos owned by King Frederick the Great.
- **The Musical Offering:** This encounter led to the composition of *The Musical Offering*, which includes a 6-voice Ricercar⁸¹.

1.3.3 Mid-18th Century Germany

Theoretical Treatises and Guidance

- **J.J. Quantz (1752):** Published *Essay of a Guide to Playing the Transverse Flute*⁸². He noted that the fortepiano provided great convenience for shading sounds⁸³ during accompaniment.
- **C.P.E. Bach (1753):** Published his seminal work, *Essay on the Right Way to Play the Clavier*⁸⁴.

⁷⁸ 扬琴/大键琴的前身之一

⁷⁹ 击弦古钢琴

⁸⁰ 早期钢琴

⁸¹ 利切卡尔（一种复杂的赋格或前奏曲风格）

⁸² 长笛演奏教程/指南

⁸³ 声音的阴暗/层次变化

⁸⁴ 键盘乐器演奏法的真谛

- **C.P.E. Bach's Verdict:** He concluded the fortepiano was useful and expressive if durable and well-constructed, though he still considered it less superior than a good clavichord⁸⁵.

Innovation in Keyboard Building

- **Christian Ernst Friederici:** A pupil of Silbermann and considered the foremost builder after him.
- **Upright-Flügel (1745):** Constructed an upright version of the piano⁸⁶.
- **Tafelklaviere (1758):** Began producing square pianos⁸⁷, which were more compact for domestic use.
- **Terminology:** By the mid-18th century, German speakers commonly adopted the name “fortepiano”.

1.3.4 Issues and Features of the Early Fortepiano

Technical Limitations and Tone

- **Expression Deficiencies:** Early models lacked subtle gradations of tone.
- **Timbre:** The sound was duller than a harpsichord and possessed less sustaining power.
- **Construction:** Featured thin wire strings latched onto a wooden frame and small hammers covered in leather.

Mechanical Controls and Stops

- **Tone Modification:** Included “Lute”, “harp”, and “piano” stops⁸⁸.
- **Una Corda:** A stop used to play on only one string⁸⁹ to change volume and color.
- **Sustain Mechanisms:** Dampers⁹⁰ controlled the sustaining of sound.
- **Levers:** Control evolved from hand levers to knee levers⁹¹, with the foot pedal⁹² arriving from England by the late 18th century.

⁸⁵ 击弦古钢琴

⁸⁶ 立式钢琴的雏形

⁸⁷ 方形钢琴

⁸⁸ 音栓（用于改变音色的装置）

⁸⁹ 单弦音栓（弱音踏板的前身）

⁹⁰ 制音器

⁹¹ 膝盖杠杆（早期的踏板形式）

⁹² 脚踏板

1.3.5 Mid-18th Century Society and Lifestyle

The Rise of the Middle Class

- **Economic Growth:** German traders flourished in cities like Hamburg, Berlin, and Leipzig.
- **Dominant Social Group:** The middle class (burghers) became the dominant group, leading to increased demand for arts and education.
- **Age of Enlightenment:** A period characterized by reasoning; by 1750, German books outnumbered Latin ones four to one.

Domestic Music and Women's Roles

- **Family Activity:** Educated burghers used music as a core family activity.
- **The Clavier in the Home:** The instrument became a standard fixture in family dwellings.
- **Gender Roles:** Playing the clavier was considered ideal for middle-class women as it was an emotional outlet, added gentleness, and demonstrated modesty.

1.3.6 Clavier after 1750 and the "Cult of Feeling"

The Sturm und Drang Movement

- **Philosophical Shift:** A reaction against the rationalism of the Enlightenment⁹³, favoring nature, humanism, and intense emotion.
- **Sturm und Drang (1760s–80s):** Named after Maximilian Klinger's 1776 play⁹⁴.
- **Pietism:** This religious movement evoked a "cult of feeling," emphasizing personal emotional experience.
- **Literature:** Sentimental poetry by Goethe and Schiller became widely popular, with many poems about the "Clavier" being set to song.

The Clavier as an Emotional Symbol

- **Ode to Joy:** Friedrich Schiller's poem (later used in Beethoven's 9th Symphony) celebrates universal brotherhood and joy.

⁹³ 启蒙运动

⁹⁴ 《狂飙突进》，德国文学与音乐中的原浪漫主义运动。

- **An mein Klavier:** Lyrics by D. Schubart (set by F. Schubert) personify the piano as a “mellow-toned” companion that shares in the player’s tears and worries⁹⁵.
- **Laura at the Piano:** Another Schiller poem describing the performer as a “sorceress” whose music commands the soul and rivals the sounds of nature.

1.3.7 Musical Activities in Leipzig

A Commercial and Cultural Hub

- **J.S. Bach’s Legacy:** Bach served as the music director of St. Thomas Church from 1723 to 1750.
- **Publishing Center:** Leipzig was the center of the book trade; by the 1760s, Breitkopf began publishing catalogues of keyboard-vocal music.
- **Music Journalism:** Breitkopf-Härtel published the *Allgemeine Musikalische Zeitung*⁹⁶.

Key Figures and Institutions

- **Johann Adam Hiller:** The first Kapellmeister⁹⁷ of the Gewandhaus and creator of the *Singspiel*⁹⁸.
- **Felix Mendelssohn:** Took over the Gewandhaus in 1835 and established the music conservatory in 1843.

1.3.8 Clavier Playing and Style after 1750

Shift Toward Amateurism

- **Expanding but Shallower:** While piano playing became more widespread, the depth of technical mastery often decreased.
- **Tuning Issues:** The art of tuning could not keep pace with the rapid sale of instruments; many players did not worry about playing in tune.
- **Simplified Notation:** Unlike professionals who used figured bass⁹⁹ and complex embellishments¹⁰⁰, amateurs preferred notes to be written out fully.

⁹⁵ 《致我的钢琴》，表达了对钢琴作为灵魂伴侣的深情告白。

⁹⁶ 《通俗音乐报》，早期的重要音乐评论期刊。

⁹⁷ 乐团指挥/乐长

⁹⁸ 歌唱剧（含有对白的德国乐剧形式）

⁹⁹ 通奏低音/数字低音

¹⁰⁰ 装饰音

- **Homophonic Style:** Musical styles moved away from polyphonic¹⁰¹ complexity toward homophonic¹⁰² textures, often using the Alberti bass¹⁰³.
- **Technical Evolution:** The abundance of scales and arpeggios in new compositions led to the increased and systematic use of the thumb.

Market Democratization: The Square Piano

- **Johannes Zumpe:** Developed the square piano¹⁰⁴ in London.
- **Accessibility:** These were smaller, had simpler mechanisms, and were significantly cheaper, making them affordable for the middle class.
- **Piano Industry:** The growth of the instrument created a tripartite industry: public performances, music publication, and manufacturing.

Professional vs. Amateur Repertoire

- **C.P.E. Bach:** Composed six sets of *Kenner und Liebhaber*¹⁰⁵ (Connoisseur and Amateur).
- **Accompanied Sonata:** Often designed for amateurs, where a keyboard was accompanied by other instruments.
- **Muzio Clementi:** His Op. 2 sonatas specifically differentiated between solo works for professionals and accompanied works for amateurs.
- **Social Status:** Amateurs felt a sense of pride and status in owning professional-grade solo sonatas.

1.4 Piano Design and Development

1.4.1 Piano Development Fundamentals

Key technical improvements focused on the action and sound production:

- **Escapement**¹⁰⁶: The mechanism allowing for a "free-flying hammer."
- **Improving Sound:**

¹⁰¹复调（多声部独立）

¹⁰²主调（旋律加简单伴奏）

¹⁰³阿尔贝蒂低音（一种分解和弦伴奏音型）

¹⁰⁴方形钢琴

¹⁰⁵《致行家与爱好者》，区分专业人士与业余玩家的作品集。

¹⁰⁶擒纵机构

- Used **thicker strings**¹⁰⁷ to create a richer tone.
- Addressed the resulting higher string tension by implementing stronger framing and bracing.
- Utilized bigger hammers and a larger **soundboard**¹⁰⁸.

1.4.2 Design Variations and Market

Pianos were designed for different roles:

- **Affordability:** Considerations for price and space (cost and target buyers).
- **Mobility:** Some designs were less movable than others.
- **Location:** Designed to fit into different specific locations.

1.4.3 The Square Piano

- Invented by J. Zumpe (German immigrant to London).
- **Characteristics:** Small, affordable, and no intermediate lever. It featured hand stops for the **damper**¹⁰⁹ and lute/buff stop.
- **Market:** Appealed to upper-class women; cost 1/3 the price of a **harpsichord**¹¹⁰.
- **History:** First recorded public solo piano performance in London by J.C. Bach (1768). Sales relied heavily on J.C. Bach's endorsement.
- Known as *Tafelklavier* (Table piano) by Germans.

1.4.4 Pantalon vs. Fortepiano

Some early square pianos are misidentified as Pantalons.

- **Pantalon**¹¹¹:
 - No dampers.
 - Non-covering wooden hammers.
 - Tone-changing stops (Muted "Soft" and "Harp").
- **Fortepiano:**

¹⁰⁷加厚琴弦

¹⁰⁸音板

¹⁰⁹制音器

¹¹⁰羽管键琴

¹¹¹潘塔隆琴

- A **chordophone**¹¹² where strings are the sounding medium.
- Played by keys and sounded by "free-flying" hammers.
- Releasing a key dampens the sound (individual damper for each note).

1.4.5 Upright Grand Piano

- European square pianos were discontinued by the 1860s (American by the end of the century), but **Uprights**¹¹³ persisted.
- **Stodart (1795)**: Patented the Upright (Cabinet) Piano with a bookcase.
- **German Designs**: Created "Giraffe" or "Pyramid" pianos.

1.4.6 Upright Piano Innovators

- **John Isaac Hawkins (1800)**: Invented the upright pianino ("portable grand"). Features included strings extending to the floor and the first use of metal bars in piano framing.
- **Robert Wornum (1811)**: Made the cottage piano with **oblique strings**¹¹⁴ and invented the upright mechanical action.

1.4.7 Unusual Designs: Vis-à-vis Ditanaklassis

- By Mathias Muller (1801).
- Features two keyboards sounding an octave apart: 4' pitch (black naturals, white sharps) and 8' pitch (white naturals, black sharps).

1.4.8 John Broadwood (English School)

- Led world production by the mid-19th century.
- **Cost**: A grand piano cost 4/5 of a skilled workman's annual salary.
- **Target**: Nobility and gentry were regular customers; middle class were "chance" customers.
- **Range Expansion**:
 - 1790s: Jan Ladislav Dussek requested 5 octaves and a fifth ($F_1 - C_6$).

¹¹²弦鸣乐器

¹¹³立式钢琴

¹¹⁴斜向琴弦

- 1810: Six octaves ($C_1 - C_7$) became popular.
- Philosophy: "Would to God we could make them like muffins" (mass production focus utilizing skilled labor without machinery).

1.4.9 Comparison: English vs. Viennese Action

- **English Action**¹¹⁵:
 - **Purpose**: Public concerts.
 - **Sound**: Bigger, louder sound.
 - **Touch**: Heavier.
 - **Mechanism**: Direct blow (no intermediate), larger hammer head.
 - **Range**: C_1 to C_7 .
- **Viennese Action**¹¹⁶ (German):
 - **Purpose**: Courtly gatherings.
 - **Sound**: Delicate and silvery.
 - **Touch**: Lighter and responsive.
 - **Mechanism**: Direct blow, smaller hammer head.
 - **Range**: F_1 to F_6 .
 - **Controls**: Knee levers instead of pedals.

1.4.10 Johann Andreas Stein

- Known for finely crafted pianos with Viennese action.
- Innovations:
 - Cross A-frame bracing to avoid wood warping.
 - Introduced knee levers (switched to pedals after 1800).
- **Pedals/Stops**: Austrian pianos could have up to 7 pedals, including the "Bassoon" (buzzing), "Moderator" (muting), "Janissary" (thumping/jingling), and **Una Corda**¹¹⁷.

¹¹⁵英式击弦机

¹¹⁶维也纳式击弦机

¹¹⁷柔音踏板/单弦

1.4.11 Sebastien Erard

- 1820s: Expanded range to 7 octaves (became standard).
- **Double Escapement**¹¹⁸: Improved the English stiff action with "Repetition" action, allowing for faster note repetition.

1.4.12 Improvement for Powerful Sound

- The Challenge: Withstanding string tension without ruining the sound with metal.
- **Alpheus Babcock (1825)**: Patented the one-piece metal frame.
- **Henri Pape (1830)**: Introduced felt-covered hammers.
- **Chickering (1840)**: First to build grand pianos with iron frames.
- **Henry Steinway (1859)**: Patented the first **cross-strung**¹¹⁹ grand piano, setting the standard for louder, richer sound.
- **Bösendorfer (1909)**: Built the Imperial Model 290 (8 octaves, 97 keys) suggested by Busoni.

1.4.13 Piano & Nobility

- **Steinway**: Associated with wealth/nobility; preferred by leading performers; made by "artisans."
- **Bösendorfer**: Produces "ultimate design" pianos and collectors' items.
- **Luxury Variations**: Blüthner Crystal Edition, Schimmel Glas (see-through piano, used on Queen Elizabeth II ocean liner).

1.4.14 David Klavins

- Critic of traditional piano design; restores old pianos.
- **Model 370 (1987)**: World's largest upright piano (370cm soundboard, spans two floors).
- **Una Corda Piano (2014)**: Open design without a cabinet; 64 individually stringed keys.

¹¹⁸双擒纵结构

¹¹⁹交叉弦列

1.4.15 Early Automated Piano

- **Mechanism:** Pneumatic action using punched paper rolls.
- **History:** "Pianista" (French) and "Pianola" (US, Edwin Votey, 1898).
- **Decline:** Caused by the growth of radio and electrical recording.
- **Purpose:** Allowed playing of pieces too difficult for humans (transcending human limitations) and capturing recordings of masters (Debussy, Rachmaninov).

1.4.16 Automated Piano in the Digital Era

- **Piano-corder:** Digital signals convert to analog via **solenoids**¹²⁰ to operate keys.
- **Yamaha Disklavier:** Includes operation for all pedals.
- **Technology:** Uses **MIDI**¹²¹ and can play from CDs, floppy disks, or computers.
- **Clavinova:** Uses "sampled sound" rather than acoustic strings.

1.5 The Revolution of Piano Culture

1.5.1 Introduction

Clementi, Haydn, Mozart, and Beethoven theoretically gather to argue who should be named the "Father of the Entertainment Industry"¹²². Each composer revolutionized musical culture centered on the piano during their time, presenting new ideas about the instrument, producing groundbreaking works, and shaping high culture as the piano's popularity surged.

1.5.2 Muzio Clementi (1752-1832): The Father of the Pianoforte

Biography and Virtuosity

Born in Rome in 1752, Clementi moved to England at age 14, where he practiced long hours daily at Peter Beckford's estate. He died in Evesham in 1832 and was granted a public interment in Westminster Abbey, with his tombstone acknowledging his fame throughout Europe as a musician and composer. In the 1760s, keyboard virtuosos¹²³ were typically children, women, foreigners, or blind individuals with abnormal musical abilities (e.g.,

¹²⁰ 螺线管

¹²¹ 乐器数字接口

¹²² 娱乐产业之父

¹²³ 键盘大师/名家

Mozart and Maria Theresia Paradis). Clementi, however, brought pure technical showmanship. He famously engaged in a musical contest with Mozart under Emperor Joseph II. It was noted that Mozart published for the amateur, while Clementi published for the "devil."

Piano Culture Revolutionist

Clementi aggressively expanded piano culture through business and teaching:

- **Manufacturing:** In 1781, he asked Broadwood for a harpsichord and piano in Paris (by the 1790s, Broadwood was making 500 pianos a year). In 1798, he partnered with Longman for piano manufacturing and music publishing.
- **Marketing:** Pianist Jan Ladislav Dussek performed on and sold Clementi's pianos.
- **Teaching:** He authored *Introduction to the Art of Playing on the Pianoforte* and *Gradus ad Parnassum* for advanced students.
- **Publishing:** He was the first composer to secure copyright through simultaneous publication in major overseas markets. While publishers like Breitkopf-Hartel ran journals (e.g., *Allgemeine musikalische Zeitung*), Clementi collaborated with editors and published the first "keepsake" albums like *Apollo's Gift*, featuring facsimiles¹²⁴ of great masters.

He pioneered the industry comprehensively: from production to marketing, concert to domestic use, manufacture to instruction, and publishing to journalism.

1.5.3 Piano Culture and Women

Social Status and Education

Arthur Loesser stated, "The history of the pianoforte and the history of the social status of women can be interpreted in terms of one another." Jane Austen's *Pride and Prejudice* (1813) defined an accomplished woman as having thorough knowledge of music, drawing, and modern languages.

- Playing the piano defined a woman's upbringing and character.
- Literacy (including musical literacy) was the foundation of upper-class female education.

¹²⁴手稿复印件

The Domestic Role of the Piano

Women playing the piano was not meant for making money. Instead, it served as a "Vehicle of Attraction" to catch the eye of the right man. The piano became a fixture women were bound to, serving both for family recreation/cultivating musical culture at home, and for a woman's own privacy and solitary moments.

1.5.4 Writing for the Piano

Instrument Evolution and Composition

Masterpieces written from 1770-1820s aligned closely with the history of piano development. There were distinct differences between Viennese and English pianos, and among makers like Walter, Schantz, and Stein. Early keyboard compositions were written for whatever instrument was at hand, but Haydn, Mozart, and Beethoven began capturing the piano's expressiveness with precise notation.

- **Amateurs vs. Professionals:** There was a divergence in music. Piano concertos were for public performances by male professionals. Solo or accompanied sonatas¹²⁵ were for domestic music-making among female amateurs (though eventually for performers too). Music for publication became more instructive and detailed.

Joseph Haydn (1732-1809): Keyboard Sonatas

Haydn composed 52 sonatas over 40 years.

- **Before 1780:** Earliest sonatas were called divertimenti or partitas¹²⁶, all including a minuet. They were simple in texture (galant style¹²⁷), lacked dynamics, and were easy pieces for students/amateurs.
- **Late 1760s-1770s:** He explored a greater range of keys. His Sonata in C Minor Hob. XVI:20 embraced the *Sturm und Drang*¹²⁸ style (dark, minor, dynamic contrast, operatic), marking an obvious departure from the harpsichord. Two works explicitly offered "fortepiano" on the title page.
- **After 1780 (28 Later Sonatas):** Haydn developed a fully realized fortepiano style. He became the "Father of the Piano Trio" (piano dominates; violin/cello double the piano). All sonatas were dedicated to women (except Esterhazy) to feature them at the keyboard.

¹²⁵ 伴奏奏鸣曲

¹²⁶ 嬉游曲或组曲

¹²⁷ 华丽风格

¹²⁸ 狂飙突进运动

- **Late London Influence:** His last three sonatas exploited the new sounds of English pianos (from his 1790s London visit). Sonata in C Major Hob XVI:50 extended the range to a high a² (exceeding 5 octaves) and explicitly called for "open pedal" (damper pedal) and "pp" (una corda or moderator).

Feature	Sonata #49 (Eb Major)	Sonata #52 (Eb Major)
Setting	Private, Intimate	Concert, Grand
Dedicatee	Marianne von Genzinger	Therese Jensen / Magdalene von Kurzbeck
Target Player	Amateur	Virtuoso
Instrument	Viennese (Schantz)	English (Broadwood)

Table 1.1: Comparison of Haydn's Late Sonatas in Eb Major

Wolfgang Amadeus Mozart: Piano Concertos

Known as the Father of the Piano Concerto, Mozart wrote 27 original piano concertos, mostly for his own public performances, though some were dedicated to specific women or featured an *ad libitum*¹²⁹ ensemble for chamber play.

- **Key Works:** Concerto No. 7 (three pianos, easier, for a countess/daughters); Concerto No. 10 (two pianos, demanding, for him and his sister).
- **Concerto No. 9 in Eb ("Jeunehomme", 1777):** Possibly for Victoire Jenamy. Featured brilliant passages making the soloist an equal partner to the orchestra, an unusual piano entrance at the beginning, two written cadenzas¹³⁰, and an unusual minuet interrupting the fast last movement.
- **Vienna Concertos (K.413-415):** A "happy medium" between too easy and too difficult. Featured figured bass notation to keep the ensemble together.
- **Grosse concerte (K.449, K.450 in 1784):** Signaled a change to fully realized symphonic textures meant to "make the performer sweat." Written for the Anton Walter fortepiano.
- **The Great Six (K.466-503, 1785-6):** Demonstrated operatic theatricality (K.466) and the elegant vocal capacity of the piano (K.488) in perfect partnership with the orchestra.

Ludwig van Beethoven (1770-1827): Piano Sonatas

Beethoven's 32 sonatas were called "the New Testament" of piano music by Hans von Bülow.

¹²⁹ 随意/即兴

¹³⁰ 华彩乐段

- **Style & Demand:** He utilized a larger range of keys (one-third in minor, compared to Mozart's 2/18 and Haydn's 5/52). He demanded virtuosity and stamina, utilizing explosive contrasts of texture, dynamics, and tempo.
- **Publication:** Continuous publication suggested high demand, ending the common practice of offering sonatas in groups. They were dedicated to both men and women (mostly aristocratic patrons) and gained nicknames from listeners (Moonlight, Pastoral, Tempest, Appassionata, Cuckoo).
- **Stretching the Instrument:** Compositions through 1802 stayed within 5 octaves. "Waldstein" and "Appassionata" stretched to $5\frac{1}{2}$ octaves, exploiting extreme registers. "Hammerklavier" required $6\frac{1}{2}$ octaves and was his longest, most technically demanding. Beethoven was acquainted with 14 different pianos (including Broadwood, Erard, Walter, Streicher, Graf).
- **Use of Pedals for Color:**
 - Piano Concerto No. 4 (End of 2nd mov): Trills pp to ff.
 - Moonlight Sonata: "sempre pp senza sordino" (use of damper and moderator).
 - Waldstein: Extended pedal markings idiomatic to Viennese pianos.
 - Sonata Op. 110 (2nd mov): Bebung-like¹³¹ repeated notes with dynamic changes and tempo fluctuation.
 - Tempest Sonata Op. 31 No. 2 (1st mov): Long pedal with pp.

1.6 Piano Lesson

1.6.1 Philosophy and Impact of Piano Study

- The study of the piano brings up debates regarding its primary purpose: whether it should focus on cultivating artistic expression or be valued as a tool for discipline and cognitive development.
- There are also pedagogical¹³² debates on whether to prioritize creativity and improvisation¹³³ over technical mastery in performance.
- Socially, questions arise about whether piano study is an elitist pursuit tied to privilege or a universal art form, and whether the music examination system (like ABRSM) functions more as a business enterprise than an educational tool.

¹³¹ 震音效果 (Bebung)

¹³² 教学法的

¹³³ 即兴演奏

- Piano lessons have profound effects on students, impacting them physically, mentally, and emotionally, while also shaping their relationships and lifestyle.

1.6.2 18th Century Keyboard Pedagogy (C.P.E. Bach)

- In the 18th century, there were no music conservatories; teaching relied on composers' letters and pedagogical¹³⁴ treatises. Lessons were often daily, with the teacher constantly beside the student.
- Keyboard playing was tied to service activities like accompanying instrumentalists, realizing basso continuo¹³⁵, and leading orchestras.
- C.P.E. Bach's *Essay on the True Art of Playing Keyboard Instruments* (1753, 1762) is a seminal work. It covers fingering (introducing the use of the thumb), embellishments¹³⁶, performance, and thorough bass¹³⁷.
- Bach favored the clavichord¹³⁸ for studying good performance and the harpsichord¹³⁹ to develop finger strength. He noted the newer pianoforte¹⁴⁰ had fine qualities, but its touch was difficult to master.
- Bach believed that keyboardists whose chief asset is mere technique are at a disadvantage; they overwhelm the hearing without satisfying it and stun the mind without moving it.

1.6.3 The Turn of the 19th Century (Milchmeyer and Clementi)

- By the end of the 18th century, the piano was replacing the harpsichord¹⁴¹, and solo piano repertory blossomed. Orchestral music stopped requiring a keyboardist.
- J.P. Milchmeyer's *The True Art of Playing the Pianoforte* (1797) introduced rounded finger positions and completely omitted harmony instruction, treating the piano as a secure solo instrument.
- Muzio Clementi's *Introduction to the Art of Playing on the Pianoforte* (1801) focused on rudimentary reading, fingerings, scales, and included 50 Preludes & Lessons. His later *Preludes & Exercises* (1811) was full of scale passages running through all keys.

¹³⁴教学法的

¹³⁵通奏低音 (一种和声伴奏形式)

¹³⁶装饰音

¹³⁷通奏低音 (一种和声伴奏形式)

¹³⁸击弦古钢琴

¹³⁹拨弦古钢琴

¹⁴⁰早期钢琴 (现代钢琴的前身)

¹⁴¹拨弦古钢琴

1.6.4 19th Century Mechanistic Ideals and Etudes

- This era saw the rise of etudes¹⁴² (finger exercises) by composers like Hanon, Czerny, and Cramer. Hanon's *The Virtuoso Pianist* aimed to build agility, strength, and perfect equality of the fingers.
- Methods advocated for an hour of daily scale practice. Friedrich Wieck strictly warned against using mechanical finger stretchers — a device that famously ruined Robert Schumann's hands.

1.6.5 Schumann and Musicality over Mechanism

- Reacting against poor-quality pedagogical¹⁴³ music and the escalating demands of middle-class hausmusik, Robert Schumann composed *Album for the Young, Op.68* (1848).
- It was written from a child's perspective to awaken imagination, specifically avoiding virtuosic display.
- Schumann's *Rules for the Musical Home and Life* advised cultivating a sense of hearing, practicing sight singing¹⁴⁴, and avoiding the brilliant popularity of virtuosos¹⁴⁵.
- He urged students not to shut themselves up like hermits practicing mechanical studies, but to seek living musical intercourse and play with heart.

1.6.6 Sound Shaping, Singing Tone, and Pedaling

- Theodor Leschetizky (1830–1915), a famous teacher in Vienna, taught legendary pupils like Paderewski, focusing heavily on producing a singing tone.
- Modern pianists like Murray Perahia emphasize translating the sounds in your head directly to your fingers, mimicking orchestral colors (like an oboe or horn) to create the “illusion” of color on the piano.
- The sustain pedal¹⁴⁶ is vital; Anton Rubinstein called it “the soul of the piano” for making a singing line possible. However, misuse can turn the sound to mud, as Friedrich Wieck complained about the “growling and buzzing” of an excessively pedaled piano.

¹⁴²练习曲

¹⁴³教学法的

¹⁴⁴视唱

¹⁴⁵炫技名家 / 演奏大师

¹⁴⁶延音踏板

Quiz

2.1 Quiz 1

In quiz 1, the focus is on the meaning of some technical terms and their usage, and also the characteristics of different kinds of pianos.

Table 2.1: Important Literature, Treatises & Poems

Title / Work	Author & Significance	Chinese Translation
<i>Giornale de' Letterati d'Italia</i> (1711)	Scipione Maffei. The journal article that first described Cristofori's piano and included diagrams.	《意大利文学学报》 (马费伊在此发表了关于钢琴发明的第一篇文章和图解)
<i>Critica Musica</i> (1725)	Johann Mattheson. Published the German translation of Maffei's article, spreading piano knowledge to Germany.	《音乐评论》(马泰松的著作, 刊登了关于钢琴发明的德语翻译)
<i>12 Sonatas for Cembalo di piano e forte</i> (1732)	Ludovico Giustini. The first music specifically published for the piano. Dedicated to Portuguese royalty.	《钢琴奏鸣曲十二首》 (朱斯蒂尼创作, 历史上第一部专门为钢琴出版的乐谱)
<i>Essay on the Right Way to Play the Clavier</i> (1753)	C.P.E. Bach. A major treatise. He praised the clavichord but admitted the piano is useful if durable.	《键盘乐器演奏法的真谛》 (C.P.E. 巴赫的重要论著, 确立了键盘演奏的规范)
<i>Guide to Playing the Transverse Flute</i> (1752)	J.J. Quantz. Noted that the piano is convenient for accompanying due to its ability to shade sound.	《长笛演奏教程》 (匡茨的著作, 提到钢琴在伴奏中表现力丰富)

Title / Work	Author & Significance	Chinese Translation
<i>Ode to Joy</i> (An die Freude)	Friedrich Schiller. Poem about brotherhood/joy, used in Beethoven's 9th Symphony. Reflects Enlightenment/Romantic ideals.	《欢乐颂》(席勒的诗作, 贝多芬第九交响曲的歌词来源)
<i>An mein Klavier</i> (To My Piano)	C.F.D. Schubart (Lyrics) / Schubert (Music). Personifies the piano as a "gentle" friend that shares tears and joy.	《致我的钢琴》(舒伯特作词, 舒伯特作曲, 将钢琴视为知己的艺术歌曲)
<i>Laura at the Piano</i>	Friedrich Schiller. Poem describing a female pianist as a "sorceress" with power over life, death, and nature.	《劳拉在弹钢琴》(席勒的诗, 描绘了女性演奏者如女巫般掌控情感的魔力)

Table 2.2: Anatomy of the Piano Mechanism (From Diagrams)

Part Name	Function	Chinese Term
Escapement (Jack)	The part that pushes the hammer up but "escapes" just before impact so the hammer can rebound freely.	擒纵器 / 顶杆 (推动琴槌并在击弦瞬间脱离的关键部件)
Back Check	Catches the hammer on its return to stop it from bouncing back up to hit the string again.	托木 / 制动器 (接住回落的琴槌, 防止二次击弦)
Damper	Felt block that lifts off the string when you play, and falls back to stop the sound when you release the key.	制音器 (止音毡, 控制声音的停止)
Hammer Shank	The thin wooden rod that holds the hammer head.	琴槌柄 (连接琴槌头的细木杆)

Part Name	Function	Chinese Term
Knuckle	A leather-covered cylinder on the shank that the jack pushes against to lift the hammer.	关节 / 轴 (顶杆与琴槌柄接触的部位)
Wippen	The complex intermediate lever assembly between the key and the hammer.	联动杠杆 / 韦佩 (连接琴键与击弦机的中间传动装置)
Capstan	A screw on the key used to adjust the height/contact with the wippen.	卡钉 (用于调节琴键与击弦机接触高度的螺丝)

Table 2.3: Social History & Musical Concepts

Concept	Description	Chinese Term
Burghers	The educated middle class (merchants, professionals) in Germany. They favored the clavichord and family music-making.	市民阶级 / 中产阶级 (受过教育的城市居民, 喜爱家庭音乐活动)
Pietism	Religious movement emphasizing "personal feeling" over strict doctrine. Favored simple, expressive music (Clavichord).	虔敬主义 (强调个人情感体验的宗教运动, 推动了音乐向情感化发展)
Sturm und Drang	"Storm and Stress" (late 1760s). Literary/musical movement focusing on extreme emotion, nature, and rebellion against rationalism.	狂飙突进运动 (强调强烈情感爆发的文学艺术运动)
Empfindsamkeit	(Implied) The "Cult of Feeling." A style of sensitivity and gentle emotion, associated with CPE Bach and the clavichord.	敏感风格 / 情感崇拜 (一种追求细腻情感表达的艺术风格)

Concept	Description	Chinese Term
Singspiel	German musical drama with spoken dialogue. Created/promoted by Johann Adam Hiller.	歌唱剧 (一种带有对白的德国歌剧形式)
Alberti Bass	A simple accompaniment pattern (broken chords) common in amateur piano music (Classical era).	阿尔贝蒂低音 (一种简单的分解和弦伴奏音型)
Bebung	Vibrato effect on the Clavichord created by varying finger pressure. Impossible on Harpsichord or Piano.	颤音 / 揉弦 (击弦古钢琴特有的手指颤音技法)

Table 2.4: Key Historical Figures (Who did what?)

Name	Role / Action	Chinese Role
Bartolomeo Cristofori	Inventor of the piano (1700). Worked for Prince Ferdinand de' Medici.	克里斯托福里 (钢琴发明者)
Gottfried Silbermann	German organ builder. Copied Cristofori's design. His pianos were played by J.S. Bach.	希尔伯曼 (德国钢琴制造先驱)
Johann Sebastian Bach	Unimpressed by Silbermann's piano in 1736, but approved of it in 1747 (Musical Offering).	J.S. 巴赫 (最初不喜欢钢琴, 后来认可)
Christian Friederici	Pupil of Silbermann. Built the "Upright-Flugel" (pyramid piano) and "Tafelklaviere" (square piano).	弗里德里希 (希尔伯曼的学生, 制造了早期立式和方形钢琴)
Johannes Zumpe	Developed the compact "Square Piano" in London, making it affordable for the middle class.	祖佩 (方形钢琴推广者, 使钢琴普及化)
Johann Adam Hiller	First Kapellmeister of Leipzig Gewandhaus. Creator of Singspiel.	希勒 (莱比锡格万特豪斯乐团首任指挥)

Name	Role / Action	Chinese Role
Pantaleon Hebenstreit	Famous Dulcimer player. His dynamic skill inspired the idea of a hammer-action keyboard.	海本斯特雷特 (扬琴大师, 其演奏启发了钢琴的发明)
Muzio Clementi	"Father of the Pianoforte" (playing technique). Composed sonatas for professionals vs amateurs.	克莱门蒂 (被称为 "钢琴之父", 区分了专业与业余作品)

Table 2.5: Comparison of Early Keyboard Instruments

Feature	Harpsichord	Clavichord	Piano (Early)
Action	Plucked (Plectrum)	Struck (Tangent remains on string)	Struck (Hammer escapes)
Dynamics	None (Terraced only via stops/manuals)	Yes (Touch sensitive) but very quiet	Yes (Touch sensitive), loud and soft
Tone	Brilliant, silvery, short sustain	Delicate, intimate, expressive	Mellow (leather hammers), singing tone
Primary Use	Orchestra, Aristocracy	Opera, Home practice, learning, "Solitude"	Chamber music, accompanying voice

2.2 Quiz 2

1. Week 5:

- (a) Square piano: Invented by J.Zumpe
- (b) Upright Grand Piano: Patented by Stodart. German made "girafe" or "pyramid" piano.
- (c) John Isaac Hawkins: Inventor of upright piano.
- (d) Robert Wornum: Invent the upright mechanical action.
- (e) Vis-a-vis ditanaklassis: Invented by Mathias Muller.
- (f) John Broadwood: Led the world production by mid-19th century.

- (g) Johann Andreas Stein: Viennese action piano.
- (h) Sebastien Erard: 7 octaves.
- (i) David Klavins: World largest upright piano

2. Week 6:

- (a) Muzio Clementi: Father of the PianoForte. Keyboard virtuoso
- (b) Joseph Haydn: Keyboard Sonatas before 1780
- (c) Mozart: Piano Concertos.
- (d) Beethoven: Piano Sonatas. Use of pedals for colors.

3. Week 7:

- (a) Leopold taught W.A. Mozart.
- (b) C.P.E. Bach: Essay on the True Art of Playing Keyboard Instruments (1753, 1762)
- (c) J.P. Milchmeyer: The True Art of Playing the PianoForte (1797)
- (d) Muzio Clementi: Introduction to the Art of Playing on the Pianoforte (1801). Preludes & Exercises.
- (e) Carl Czerny: Letters to a Young Lady on the Art of Playing Piano (1842), Forty Daily Exercises. Regularity, evenness, exactness brought the mechanistic ideals.
- (f) Etudes (Finger exercises): Czerny, Cramer, Clementi, Chopin, Hummel, Pischna, Brahms, etc.
- (g) Hanon: The Virtuoso Pianist.
- (h) D.G. Turk's Klavierschule: Suggested amateur to practice a handful of hours per week.
- (i) Hunter: Instructions for the pianoforte.
- (j) Herz: Complete method of piano.
- (k) Frederich Wieck: Recommended quarter of an hour daily of scales.
- (l) R.Schumann: Used a finger-strengthening device that ruined his hands. Album for the Young.
 - Spring Song
 - Gathering of Grapes-Happy Time

- Haverst Song
- Wintertime

(m) Theodor Leschetizky: Beethoven → Czerny → Liszt, Leschetizky

4. Week 8

- (a) Jan Ladislav Dussek: Receptient of Broadwood 6-octaves piano.
- (b) Broadwood and Clementi, Clementi company and Dussek, Pleyel and Kalkbrenner, Erard and Liszt
- (c) Thalberg-Liszt concert organized by Belgiojoso.
 - Liszt: Pacini's Niobe. Piano transcription on Berlioz's Symphonie fantastique.
 - Thalberg: Fantasia on Rossini's Moses. Art of Song Applied to the Piano.
- (d) Schubert performed his songs, Winterreise
- (e) Chopin: dance pieces.
- (f) The Religion of the Pianoforte by George Bernard Shaw: The pianoforte is the most important of all musical instruments; its invention was to music what the invention of printing was to poetry
- (g) The Wonder of the Piano by Menahem Pressler: He found a very good piano that he liked.
- (h) On Audiences by Vladimir Horowitz: The concentration of the artist is contagious to the public.
- (i) Girl at the Piano by Paul Cezanne.
- (j) Pierre Auguste Renoir: The most prolific painter of woman at the piano. He painted Young girls at the Piano (1892) Yvonne and Christine Lerolle at the Piano (1897).

Figure	Representative Works (曲目/著作)	Characteristics & Style (特色)
Muzio Clementi	• Introduction to the Art of Playing on the Pianoforte • Gradus ad Parnassum	• Known as "The Father of the Pianoforte". • Offered high technical showmanship and composed primarily for virtuosos (the "devil") rather than amateurs.

Figure	Representative Works (曲目/著作)	Characteristics & Style (特色)
Joseph Haydn	• Early: Sonata in C Minor Hob. XVI:20 • Late: Sonata in C Major Hob. XVI:50	• Known as the “Father of Piano Trio”. • Early works were simple and in the galant style, while later works explored Sturm und Drang. • Later sonatas stretched the fortepiano register and used pedal markings such as <i>una corda</i> and open pedal.
W. A. Mozart	• Concerto No. 9 in E$\flat$ “Jeune-homme” • Concertos K.449, 450 (“Grosse concerti”) • Concertos K.466 and K.488	• Known as the “Father of the Piano Concerto”. • Unlike Clementi, he often published for amateurs. • His concertos combined brilliant passages, symphonic textures, and partnership between piano and orchestra.
Ludwig van Beethoven	• 32 Piano Sonatas (the “New Testament” of piano) • “Waldstein” Sonata • “Appassionata” Sonata • “Hammerklavier” Sonata	• Stretched the mechanical and expressive capabilities of the piano to the limit. • Music demanded immense virtuosity and stamina. • Known for explosive contrasts in texture and dynamics and extreme tempos.
C. P. E. Bach	• Essay on the True Art of Playing Keyboard Instruments	• Believed true keyboard artistry depended on fingering, embellishments, and expressive performance. • Players relying only on technique are at a disadvantage.

Figure	Representative Works (曲目/著作)	Characteristics & Style (特色)
Carl Czerny	- Letters to a Young Lady on the Art of Playing Piano - Various Etudes (Finger Exercises)	- Promoted mechanistic ideals in piano playing. - Emphasized regularity, evenness, and striking the keys perpendicularly. - Warned against rushing the tempo, calling fingers “disobedient creatures”.
Robert Schumann	- Album for the Young, Op.68 (e.g., The Happy Farmer, Wild Rider) - Rules for the Musical Home and Life	- Wrote pedagogical pieces from a child’s perspective to inspire imagination. - Believed players must have music in their “head and heart,” not just in their fingers.
Jan Ladislav Dussek	N/A (Noted as a Czech virtuoso)	Historically significant as the first performer to sit sideways to the audience during a piano performance.
Franz Liszt	Transcriptions and paraphrases (e.g., Berlioz’s <i>Symphonie fantastique</i>, Pacini’s <i>Niobe</i>)	- Legendary virtuoso and first pianist to employ an agent to manage concert tours. - Famous for dramatic piano transcriptions of orchestral and operatic works.
Sigismond Thalberg	- Art of Song Applied to the Piano, Op.70 - Fantasia on Rossini’s <i>Moses</i>	- Famous for piano fantasies based on operatic melodies. - Produced influential arrangements of songs and operas for piano.
Charles-Louis Hanon	The Virtuoso Pianist	- Focused on agility, independence, strength, and equality of the fingers. - His exercises were described by Rachmaninov as the “secret of the Russian school”.

Figure	Representative Works (曲目/著作)	Characteristics & Style (特色)
Theodor Leschetizky	The Groundwork of the Leschetizky Method	- Known as the “virtuoso teacher of virtuosos”. - Emphasized singing tone and high-finger technique.
Franz Schubert	Winterreise	- Often accompanied himself while singing. - Wrote dance music in short repetitive segments suitable for parlor dancing.
Frederic Chopin	Various dance pieces	Wrote idealized dance pieces intended for concert listening rather than actual dancing.

2.3 Quiz 3

Table 2.7: Comprehensive Summary of Concert History and Virtuosos (SHP Weeks 9 & 10)

Topic / Figure	Key Characteristics & Historical Context	Notable Facts / Repertoire
Early Concert Culture (London & Vienna)	London: Evolved from tavern concerts and <i>collegia musica</i>¹. Broadwood pianos gained dominance. Vienna: Mozart played in private aristocratic homes (Esterhazy, Galitzin) and converted warehouses before dedicated halls emerged.	London: Hanover Square Rooms (1775); J.C. Bach used a fortepiano² (1767). Vienna: Musikverein (1870); Society for the Friends of Music (1812).

Continued on next page

¹音乐社团
²早期钢琴/古钢琴

Table 2.7 – continued from previous page

Topic / Figure	Key Characteristics & Historical Context	Notable Facts / Repertoire
Early Concert Culture (Paris & NY)	Paris: Piano manufacturers' showrooms (Erard, Pleyel) became top venues for virtuosos³ to build reputations. New York: Concert halls built by piano makers (Steinway, Chickering) heavily influenced the rise of virtuoso tours.	Paris: Concert Spirituel (1725-90); Paris Conservatory Hall (1811). NY/USA: Sigismond Thalberg (320 concerts); Carnegie Hall opened with Tchaikovsky (1891).
The Solo Recital⁴	Concerts used to feature mixed programs (symphonies, chamber music ⁵ , singers). Liszt famously reduced performers until he declared, "I am the concert."	1840: Hanover Square Rooms first used the word "Recitals" for Liszt. 1870s: Clara Schumann adopted the format.
Franz Liszt (1811-1886)	Considered the greatest contemporary pianist. Sparked "Lisztomania" (fans fought over his gloves, hair, and broken piano strings).	Known for absolute stage charisma, transcendent technique, and deep communication with the audience.
Russian Conservatories (Est. 1860s)	St. Petersburg (1862): Introspective, focused on structure, played looking at the keys without pedal. Moscow (1866): Outgoing, focused on color, virtuosity, lots of pedal, and grand gestures.	St. Petersburg: Founded by A. Rubinstein. Grads: Tchaikovsky, Prokofiev. Moscow: Founded by N. Rubinstein. Grads: Rachmaninoff, Scriabin.
Anton Rubinstein (1829-1894)	Liszt called him "Van II". Likened to a biblical Samson. Endured a grueling US tour (215 concerts in 239 days), which he despised as "slavery."	Created massive programs (1885-86) designed to exhibit the entire scope of piano literature (from Byrd to Liszt).
Vladimir Horowitz (1903-1989)	"The 20th Century Liszt". Thunderous sound, contrast of tone colors. Unusual technique: flat fingers, palms below keys, immobile body.	Dazzling Tchaikovsky 1st Concerto ⁶ ; revived interest in Scarlatti and Clementi sonatas ⁷ ; toured with his own piano and technician.
Continued on next page		

³炫技大师/名家⁴独奏会⁵室内乐⁶协奏曲⁷奏鸣曲

Table 2.7 – continued from previous page

Topic / Figure	Key Characteristics & Historical Context	Notable Facts / Repertoire
Sergei Rachmaninoff (1873-1943)	Expressive, song-like melodies. Played with a “dry ice” coolness (cool so that it’s hot). Openly criticized Horowitz’s playing as un-musical.	24 Preludes ⁸ (covering all keys); Piano Concerto No. 2 (spawned pop hits like “All by Myself”); Paganini Variations.
Hans von Bülow (1830-1894)	German virtuoso and conductor. Liszt’s student. Like A. Rubinstein, he hated his exhausting 1875 US tour (sponsored by Chickering).	First to perform all Beethoven sonatas from memory. Premiered Liszt’s B minor Sonata and Tchaikovsky’s 1st Concerto.
Arthur Schnabel (1882-1951)	Prioritized intellectual insight and artistic truth over technical bravura ⁹ . Believed the pianist is just a “mountain guide” for the composer’s work.	Mastery of Schubert and late Beethoven. First to record all Beethoven sonatas (dubbed “the man who invented Beethoven”).
Leopold Godowsky (1870-1938)	Combined Russian sparkle with Germanic precision. Had small hands but achieved incredibly light, logical, and super-acrobatic playing.	Wrote 53 studies on Chopin’s Etudes ¹⁰ , raising their difficulty to near-impossible levels (left hand alone, interweaving).
Arthur Rubinstein (1887-1982)	The greatest Chopin interpreter. Played with serious, tender nobility (masculine Chopin). The “yin” to Horowitz’s “yang”.	Believed in not over-practicing (max 3 hours). Played the Polish anthem at the 1945 UN inauguration.
Van Cliburn (1934-2013)	Texan pianist with a rich, round tone and singing-voice phrasing. Described his fame as a “sensation,” not just success.	Won the 1st Tchaikovsky Competition in Moscow (1958) during the Cold War. Played for every US President from Truman to Obama.
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⁸前奏曲⁹华丽技巧/炫技¹⁰练习曲

Table 2.7 – continued from previous page

Topic / Figure	Key Characteristics & Historical Context	Notable Facts / Repertoire
Piano Competitions	Used to obtain fame and establish a career. Top competitions: Chopin, Tchaikovsky, Van Cliburn, Queen Elisabeth, Rubinstein.	The 19th Chopin Competition saw an “Asian wave,” with all top six prizes taken by pianists of Asian heritage (e.g., Eric Lu).
Valentino Liberace (1919-1987)	“Mr. Showmanship”. Transitioned from classical (played Liszt at 20) to highest-paid pop-culture superstar in Vegas.	“I don’t give concerts, I put up a show.” Added sentimentality and glamor to classical tunes.
Lang Lang (b. 1982)	The highest-paid classical pianist today. Known for playing at massive global events.	2008 Beijing Olympics; UN Messenger of Peace; played Gershwin at the Grammys.

Table 2.8: Comprehensive Summary of Feminine & Masculine Expression in Piano Music (SHP Weeks 11 & 12)

Composer	Characteristics	Notable Works
Franz Schubert (1797-1828)	Known as a great melodist. Described as a “fountain of good tunes” but failed in technical skill for their development. His emotions were tempered by classical restraint and never violent. To understand his piano music, one must approach it through his songs and make the piano “sing”.	Composed 600 songs, Impromptus ¹¹ , Moments Musical ¹² , and Piano Sonatas ¹³ . <i>Sonata in Bb, D.960</i> (his last sonata).
Robert Schumann (1810-1856)	A “dreamer” who suffered nervous breakdowns and hallucinations. Invented two imaginary characters to represent his split nature: Florestan (passionate/carefree) and Eusebius (passive/introvert). Frequently embedded secret codes in his works. His music is wild, impulsive, and extremely romantic.	<i>Abegg Variations Op. 1</i> (spelling A-Bb-E-G-G). <i>Carnaval, Op. 9</i> (features codes like Asch and Scha). <i>Davidbundlertanz, Op. 6</i> (expresses passionate love for Clara).
Continued on next page		

¹¹即兴曲¹²音乐瞬间¹³奏鸣曲

Table 2.8 – continued from previous page

Composer	Characteristics	Notable Works
Johannes Brahms (1833-1897)	Music was described as serious, thick, with rumbling basses, and radiating bigness. Used Classical structures but wrote in highly Romantic styles. Wrote “pure music” rather than program music ¹⁴ . Critics sometimes called his work academic and unemotional.	Early works: Large scale, highly difficult, e.g., <i>Variations on a theme by Paganini</i> . Late works: Short, introspective pieces like <i>Intermezzo Op.118 No.2</i> (heartwarming) and <i>Capriccio Op.116 No.3</i> (energetic).
Frédéric Chopin (1810-1849)	Polish composer who preferred intimate salon surroundings over large concert halls. Hated piano pounding, comparing it to a barking dog. His music is full of poetic refinement without ferocity. Deeply influenced by Bellini’s bel canto ¹⁵ arias. Preferred Pleyel pianos for their silvery sonority.	<i>Nocturnes</i> ¹⁶ (popularized from John Field’s works). <i>Mazurkas & Polonaises</i> ¹⁷ (aroused nationalistic sentiments, shifted accents to weak beats).
Claude Debussy (1862-1918)	Impressionistic ¹⁸ style; drifted away from convention to bring out mysteriousness. Piano played without hammers to provide an ethereal, weightless sound. Used harmony as a “secret weapon” for musical color and perfume. Relied heavily on soft dynamics and murmuring sounds like a glissando ¹⁹ .	Early: <i>Arabesques</i> , <i>Suite Bergamasque</i> , <i>Pour le Piano</i> . Later: <i>Estampes</i> , <i>Images</i> , <i>24 Preludes</i> ²⁰ (titles placed at the end, e.g., <i>The Girl with the Flaxen Hair</i> , <i>Mists</i>).
Ludwig van Beethoven (1770-1827)	Points to the future and radiates human struggle. Early period: quirky, abrupt, bridged salon and concert hall. Middle (Heroic) period: explored piano capabilities (pedal, extended range), abrupt modulations ²¹ . Late period: intellectual depth, formal innovation, unconventional.	<i>Pathétique Op. 13</i> . <i>Moonlight</i> , <i>Tempest</i> , <i>Waldstein</i> , <i>Appassionata</i> , <i>Les Adieux</i> . <i>Hammerklavier Op. 106</i> (longest and most technically challenging, 50 mins).
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¹⁴标题音乐¹⁵美声唱法¹⁶夜曲¹⁷波兰舞曲¹⁸印象派¹⁹滑音²⁰前奏曲²¹转调

Table 2.8 – continued from previous page

Composer	Characteristics	Notable Works
Franz Liszt (1811-1886)	Described as a “Greek god descended to earth”. Vowed to be Paganini at the piano. His playing was bombastic, searing, jaw-dropping, and fiery. Known for extreme animation and heroic fire.	<i>Six Paganini Etudes</i> ²² . <i>Twelve Transcendental Etudes</i> (dedicated to Czerny). <i>19 Hungarian Rhapsodies</i> ²³ (based on folk tunes; No. 2 featured in “Cat Concerto”).
Béla Bartók (1881-1945)	Incorporated folk elements (Hungarian pentatonic ²⁴ & Romanian chromatic). Highly rhythmic-driven, dance-like, filled with accents.	<i>Allegro Barbaro</i> (1911; his most famous solo piece).
Sergei Prokofiev (1891-1953)	Known for his most dissonant music and cluster chords. Irony of receiving the Stalin Prize while expressing true feelings after the death of the Meyerholds.	<i>Sonata No. 7 “Stalingrad”</i> (one of the War Sonatas). The 3rd movement, <i>Precipitato</i> , is a toccata ²⁵ bringing Rock & Roll explosiveness.
Alexander Scriabin (1872-1915)	Used mystic chords, leaps, tremolos ²⁶ , and percussive sounds. Reflected the accumulation of heat causing earth’s destruction. Seriously damaged his hand practicing Balakirev’s <i>Islamey</i> .	<i>Vers la Flamme</i> (Toward the Flame; originally intended as Sonata No. 11).
Mily Balakirev (1837-1910)	Composed notoriously difficult oriental fantasies. Ravel intentionally wrote <i>Gaspard de la nuit</i> to be even more difficult than Balakirev’s work.	<i>Islamey: Oriental Fantasy</i> (based on a Caucasus folk dance tune).
Igor Stravinsky (1882-1971)	Did not intend to reproduce orchestral sounds on piano, but rather display pianistic challenges. Characterized by wild jumps, polyrhythms ²⁷ , fast scales, glissandos, and tremolos.	<i>Three movements from Petrouchka</i> (ballet music written for solo piano to influence Arthur Rubinstein).

²²练习曲²³狂想曲²⁴五声音阶²⁵托卡塔²⁶震音²⁷多重节奏